

How to Make Memorable Plots

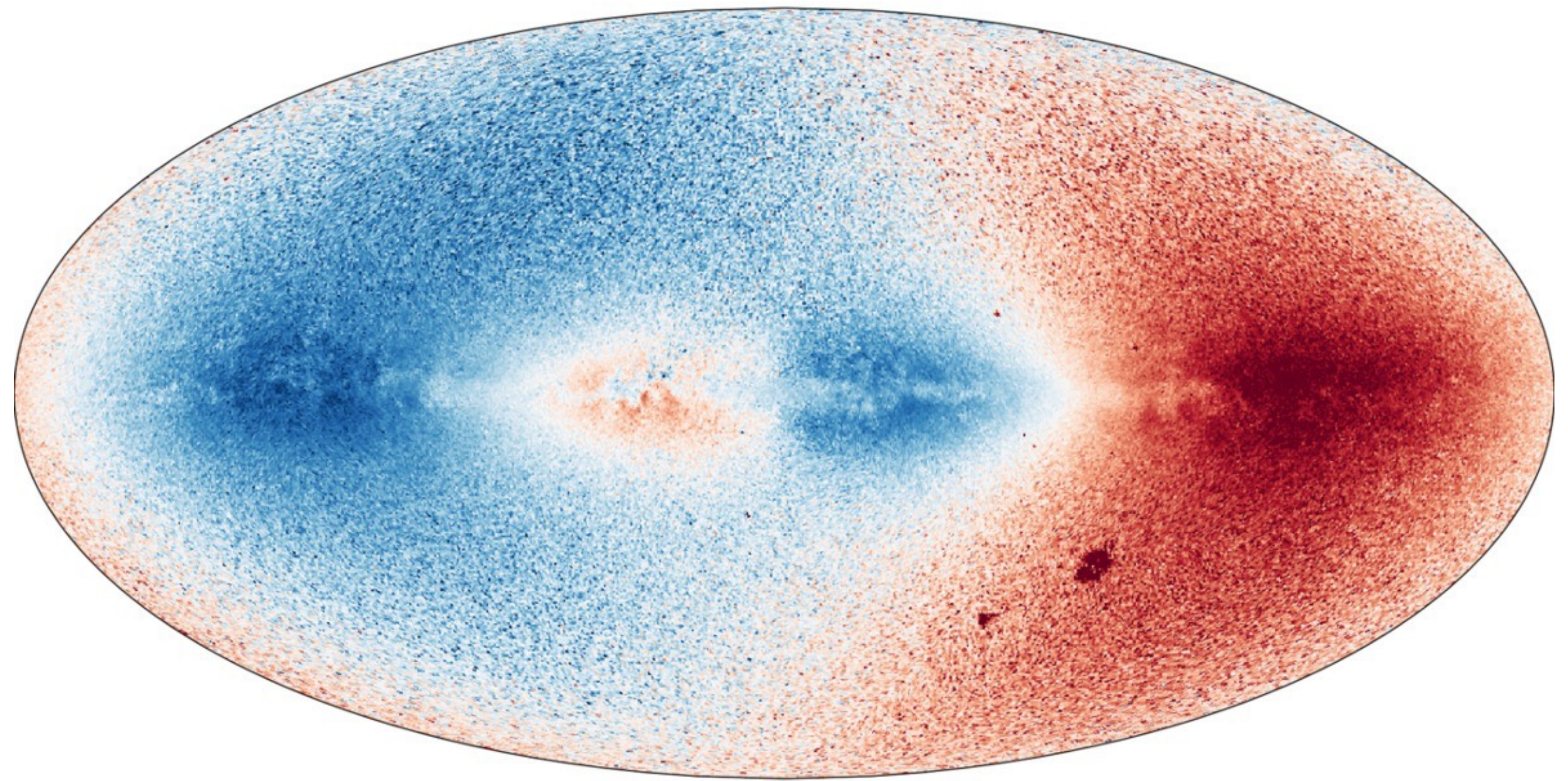
Make the science easy to **see, understand, and remember** for
scientific papers • presentations • outreach

Sven Buder, 18 August 2026, ASTR8030 & RSAA Skillshare Sessions

What makes a plot memorable?

Look at these figure for a few seconds.
Then ask yourself:

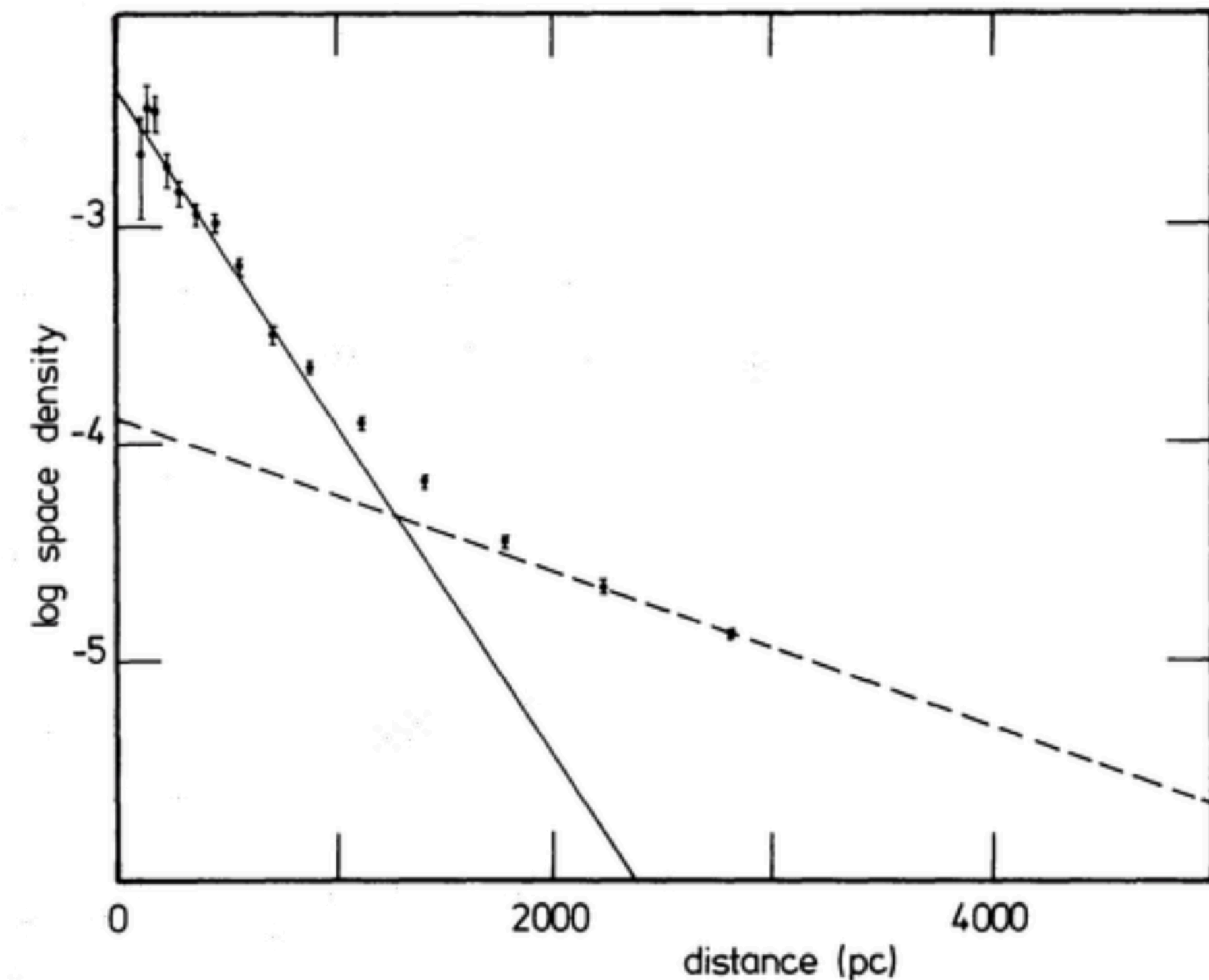
- * What did I look at first?
- * What do I remember?
- * What do I think the main message was?



What makes a plot memorable?

Look at these figure for a few seconds.
Then ask yourself:

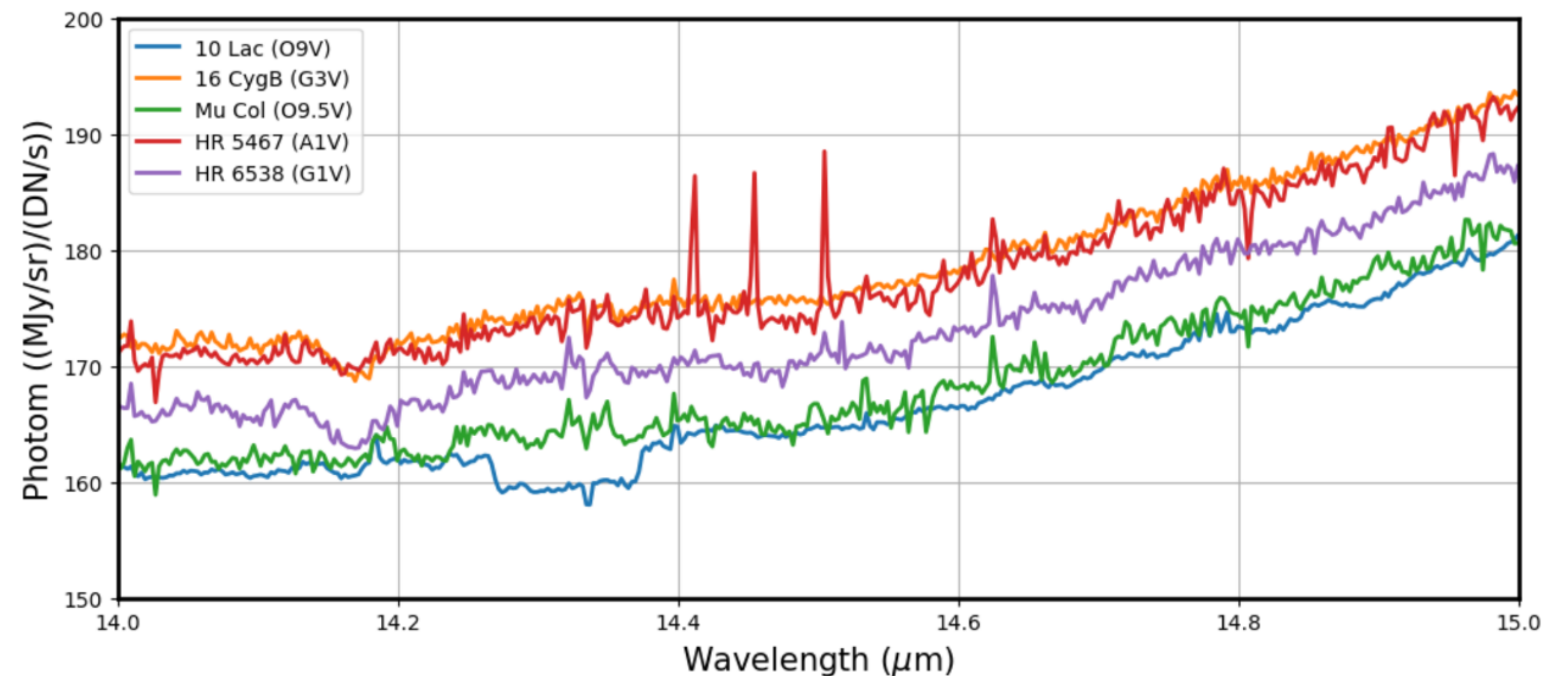
- * What did I look at first?
- * What do I remember?
- * What do I think the main message was?



What makes a plot memorable?

Look at these figure for a few seconds.
Then ask yourself:

- * What did I look at first?
- * What do I remember?
- * What do I think the main message was?



Memorable $\neq$ most beautiful and breathtaking

Three questions for every figure

What should they **SEE**?

What should they **UNDERSTAND**?

What should they **REMEMBER**?

Everything else should support those goals.

Exercise 1 — Figure memory test (2 people, 2x10 min)

1. **Person A:** Give only enough context to understand the topic/axes.
2. **Person B:** Look at the figure for **20 seconds**. No explanation yet.
3. Hide the figure.
4. Person B writes:
 - * What did I look at first?
 - * What was the main result?
 - * What do I remember?
 - * What was hard to decode?
5. Person B **draws the figure from memory**.
6. Reveal the original and compare.
7. Person A explains the intended message.
8. Discuss: **Did the visual hierarchy support the science?**

Then swap roles.

What survived?

Discuss with your partner

- * What did you remember?
- * What disappeared?
- * What attracted your attention first?
- * Was that what the author intended?
- * What would you simplify?
- * What would you emphasise?

Rule 1 — Decide what wins attention

Every element competes for attention.

If your audience notices only one thing what should it be?

Use visual hierarchy deliberately:

contrast • colour • opacity • size • line width • position • annotation

Rule 2 — Remove before you add

A common question: **How can I make this plot more impressive?**

Try this first: **What can I remove?**

Less competition

→ clearer message

→ stronger figure

Simple does not mean boring.

Rule 3 — Make it as simple as the message allows

Some science is genuinely complex.

That is fine.

The problem is not complexity.

The problem is **unmanaged complexity.**

If it has to be complex...

...build it up!

Instead of showing everything at once:

1. Start with the axes / basic idea
2. Add the main data
3. Add the model or data
4. Add the data or model
5. Highlight the interesting feature
6. Annotate the conclusion



Let the audience follow the argument.

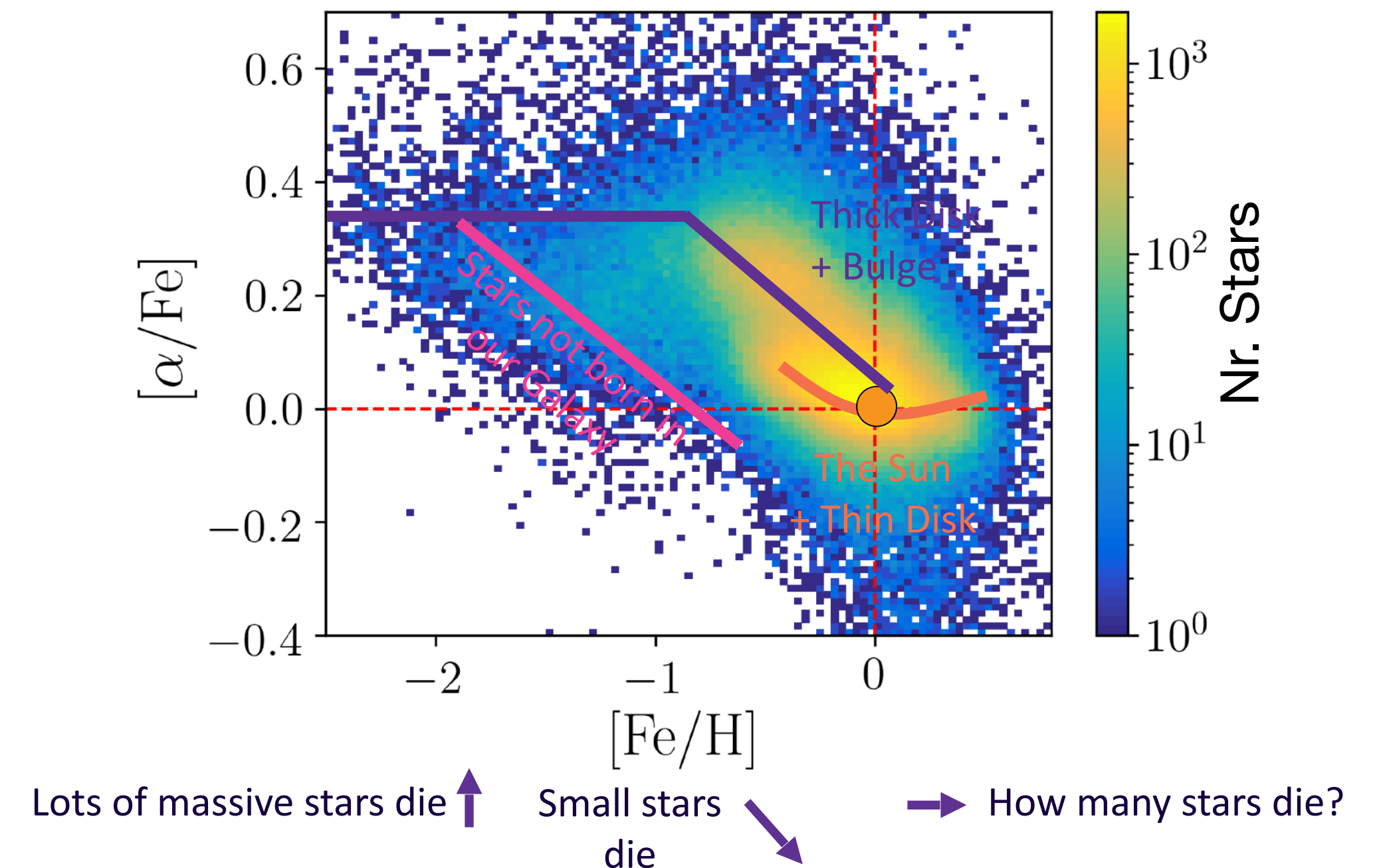
If it has to be complex...

...build it up!

Instead of showing everything at once:

1. Start with the axes / basic idea
2. Add the main data
3. Add the model or data
4. Add the data or model
5. Highlight the interesting feature
6. Annotate the conclusion

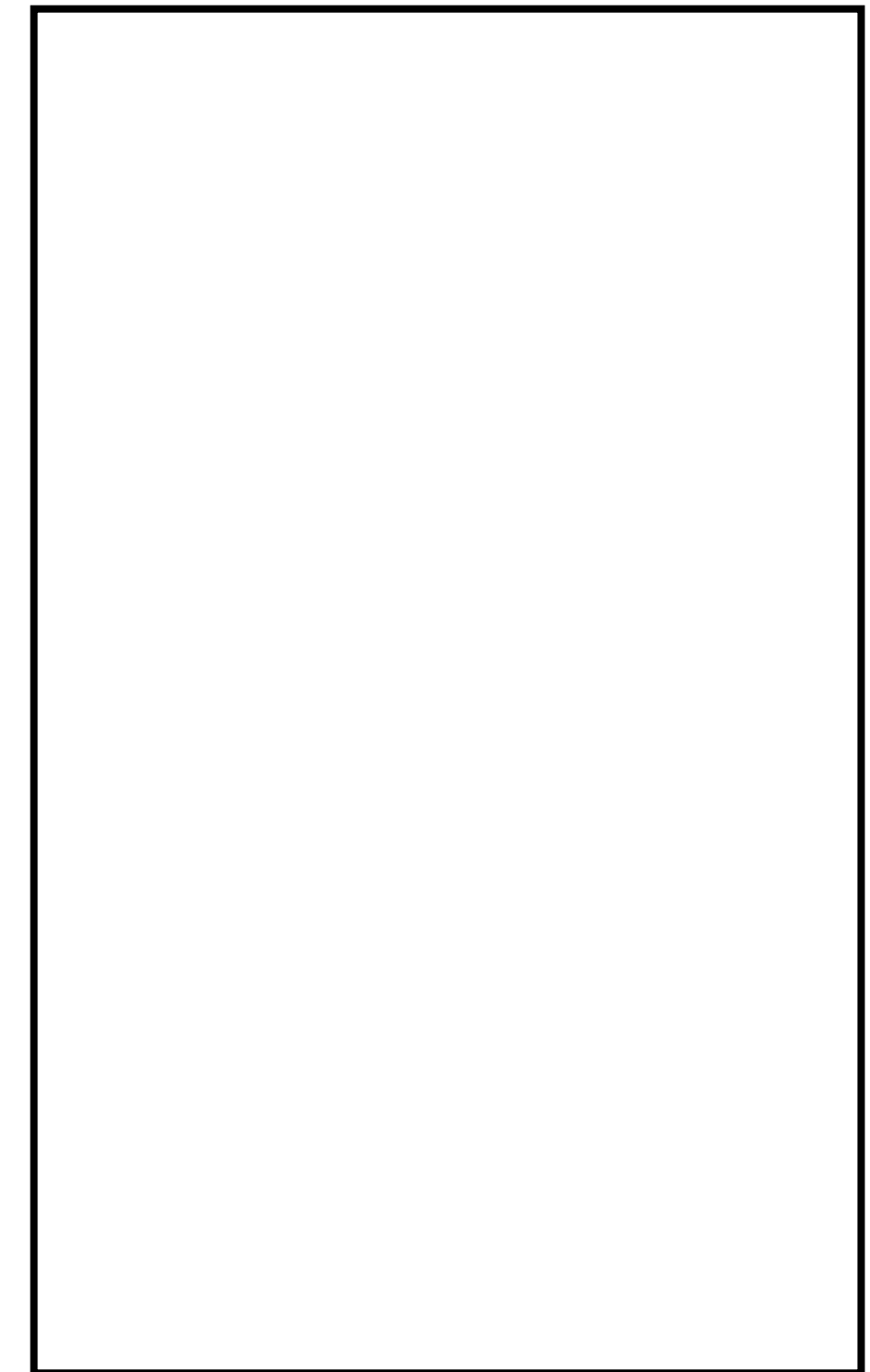
Let the audience follow the argument.



Animation can help

Animation is useful when it controls **when** information appears:

- * reveal populations one by one
- * build up a complex figure
- * zoom into an important region
- * show evolution with time
- * transition from data → interpretation



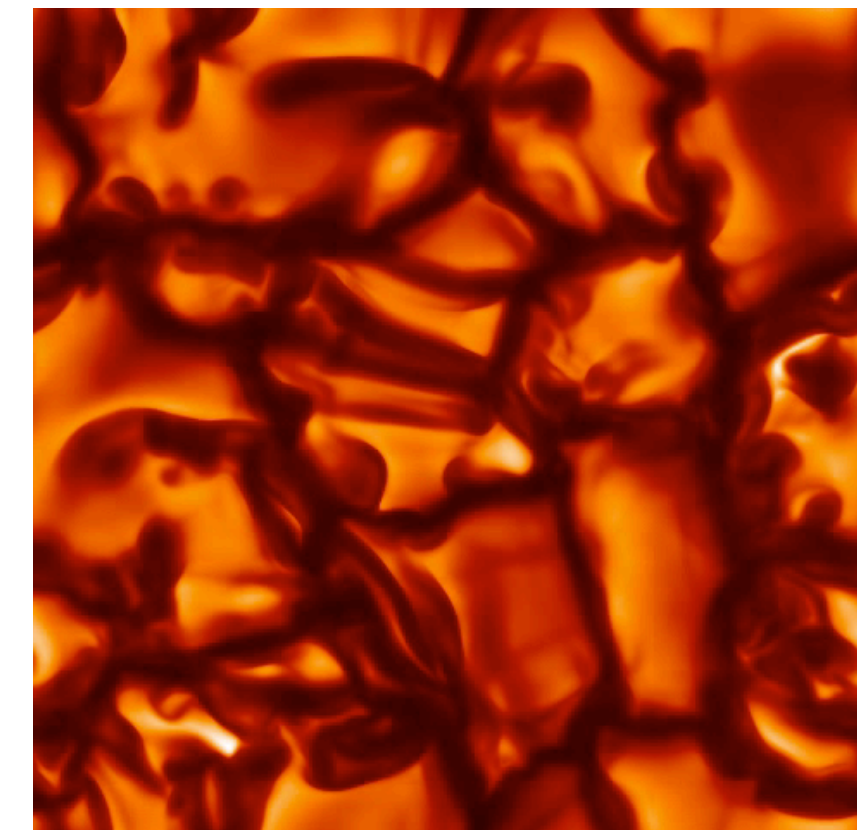
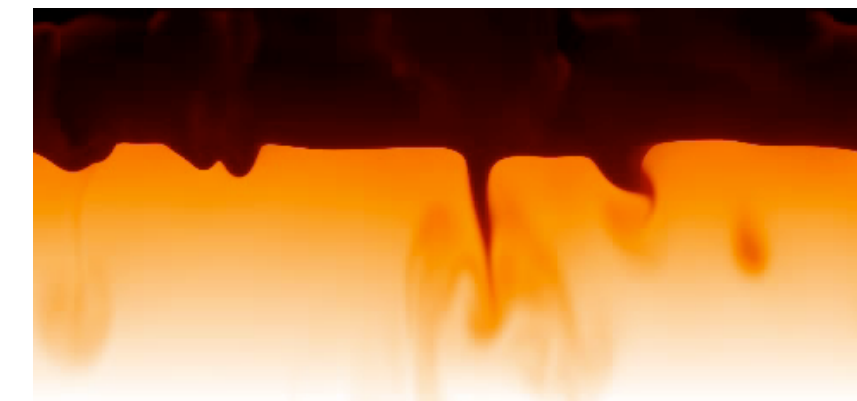
Animation should reveal the science...

...not decorate the slide.

Animation can help

Animation is useful when it controls **when** information appears:

- * reveal populations one by one
- * build up a complex figure
- * zoom into an important region
- * show evolution with time
- * transition from data → interpretation



Animation should reveal the science...

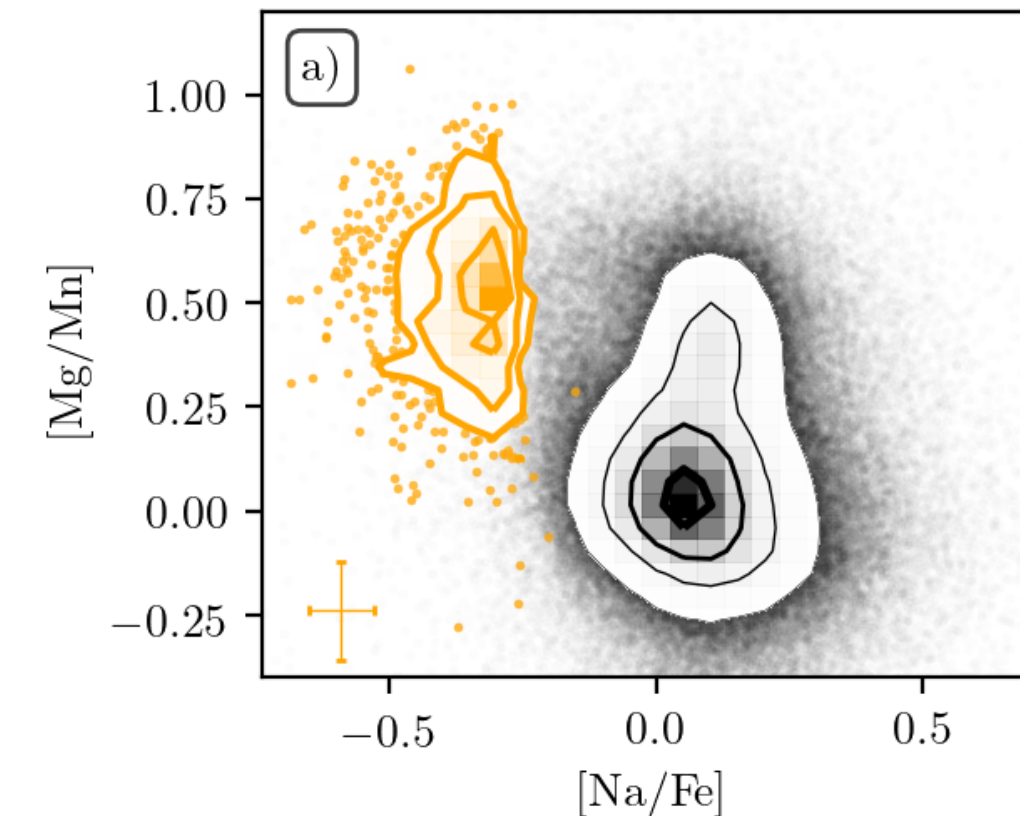
...not decorate the slide.

Rule 4 — Design for your audience

The same result may need very different figures.

Paper:

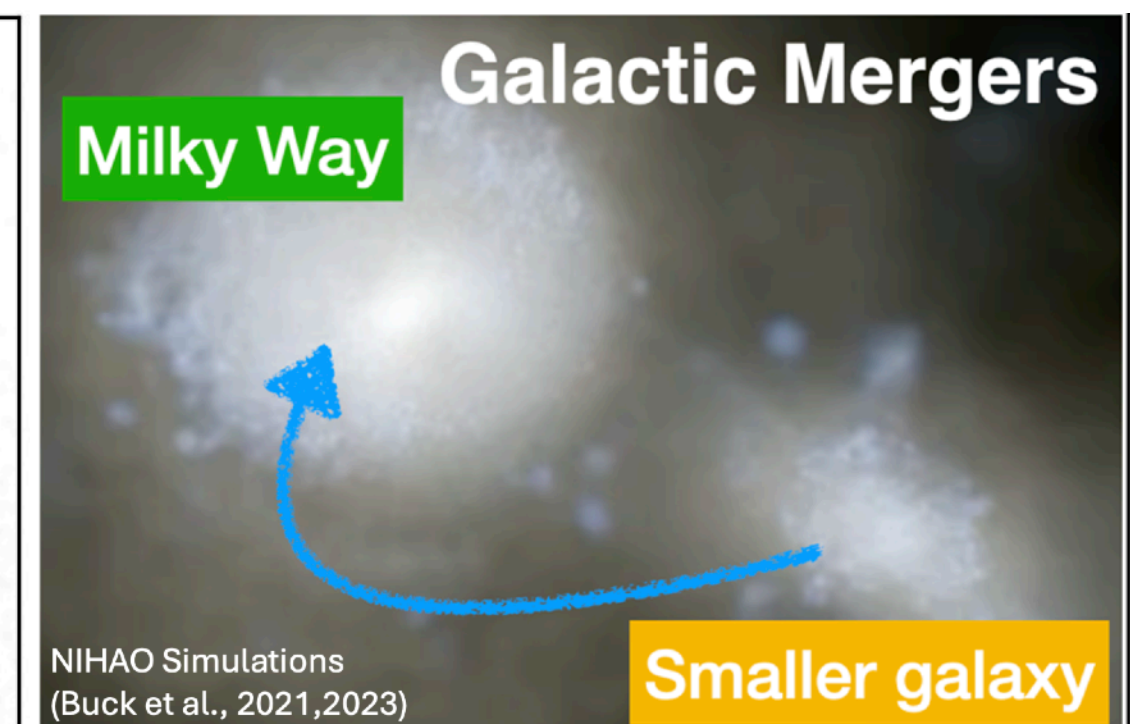
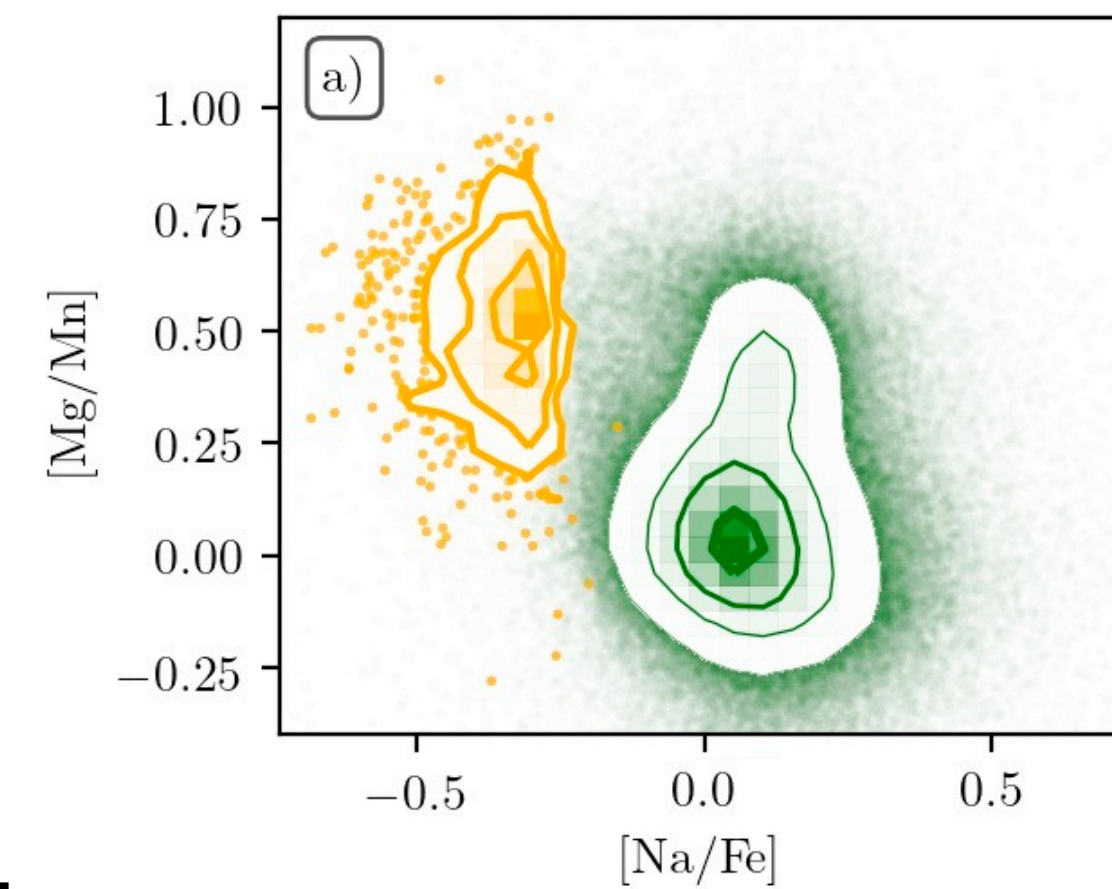
Detailed • quantitative • self-contained



Buder et al. (2022)

Specialist talk:

Fast • focused • annotated • layered



General astronomy audience:

Less jargon • more guidance • simpler context

Outreach:

Intuitive • physical • visual • minimal technical notation

Exercise 2 — One plot, different audiences

~15 minutes • groups of 3–5

Your group gets **one audience**:

I	Public outreach	II	General astronomy audience
III	Journal article	IV	Specialist conference talk

You have **5 minutes** to redesign the same figure.

Discuss:

- * What can your audience already know?
- * What is the **ONE message** they should take away?
- * What would you **remove or simplify**?
- * What would you **emphasise, annotate, or build up**?

Be ready to explain your choices in ~1 minute.

Sometimes the best redesign is...

...not using the original plot at all.

A great paper figure may be a terrible outreach or grant application figure.

A great outreach figure may not be publishable

A detailed result may become:

- one highlighted trend
- a simplified diagram
- an image + annotation
- a sequence of builds
- a completely different visual representation

Adapt the communication, not just the font size.

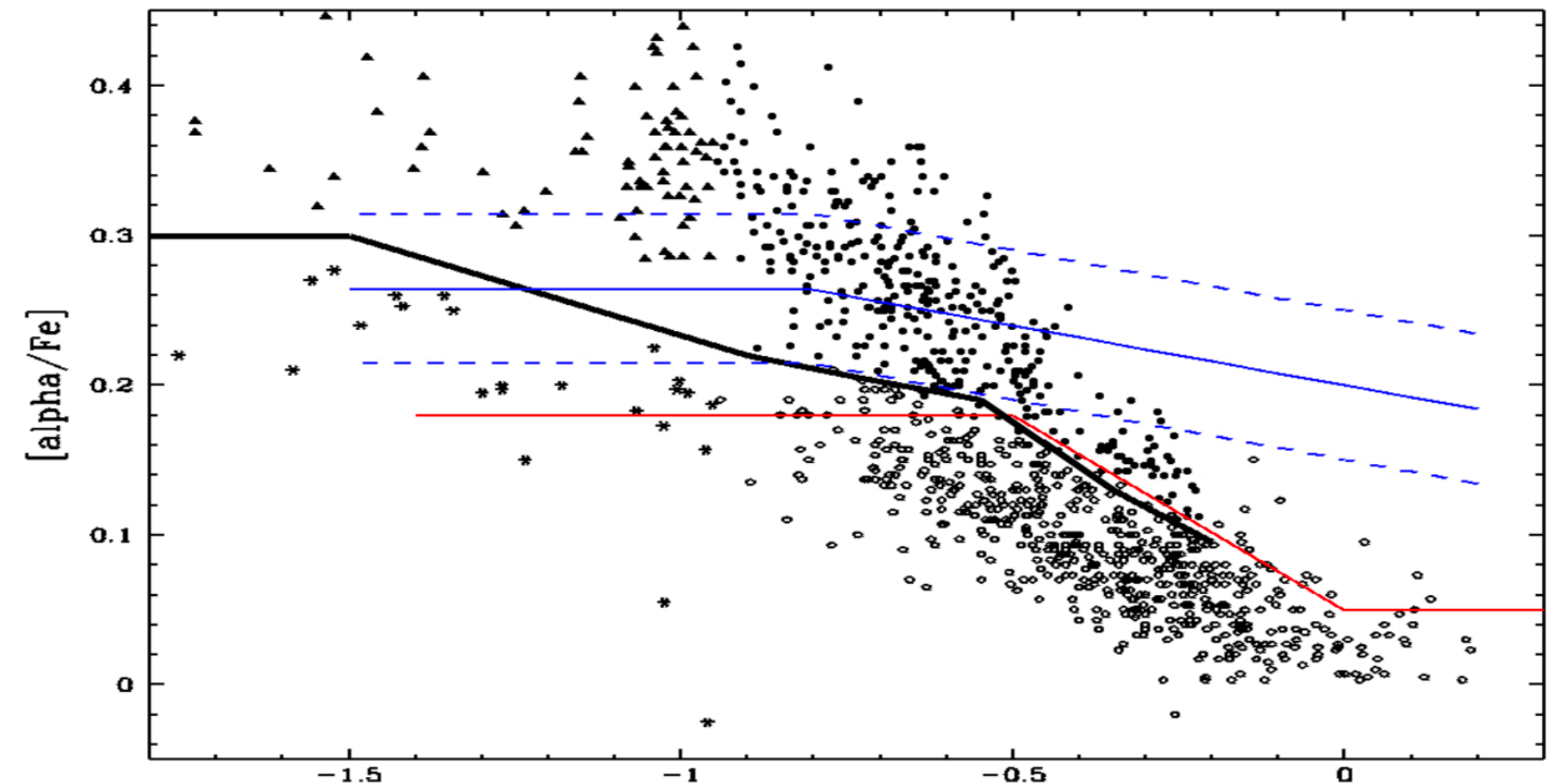
Rule 5 — A paper figure must work without you

In a presentation: **You are the caption.**

In a paper, **you are not there.**

A good paper figure should make clear:

- What is plotted?
- What are the units?
- What do colours/symbols mean?
- What uncertainties matter?
- What comparison is being made?
- What should the reader conclude?



Wow ≠ flashy

A memorable figure can create different kinds of impact.

Visual wow: A striking image or animation

Conceptual wow: A very simple plot with a surprising result

Data wow: A huge dataset revealing structure

Reveal wow: Complexity gradually resolving into an answer

The visual effect should strengthen the science.

Beautiful $\neq$ effective

A plot can be:

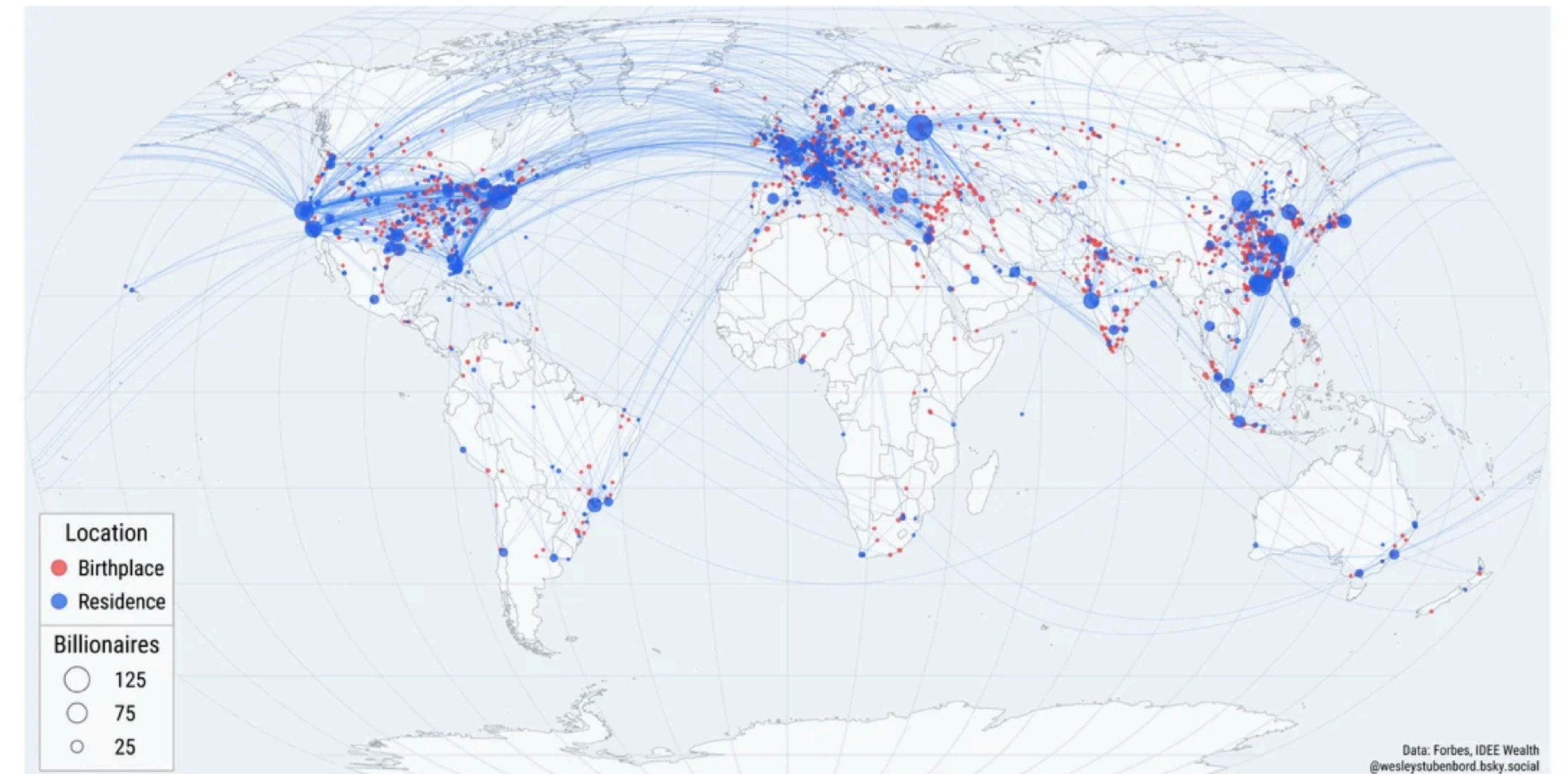
**beautiful but
confusing/complex**

or

**ugly but
informative**

Aim for:

clear + honest + visually compelling

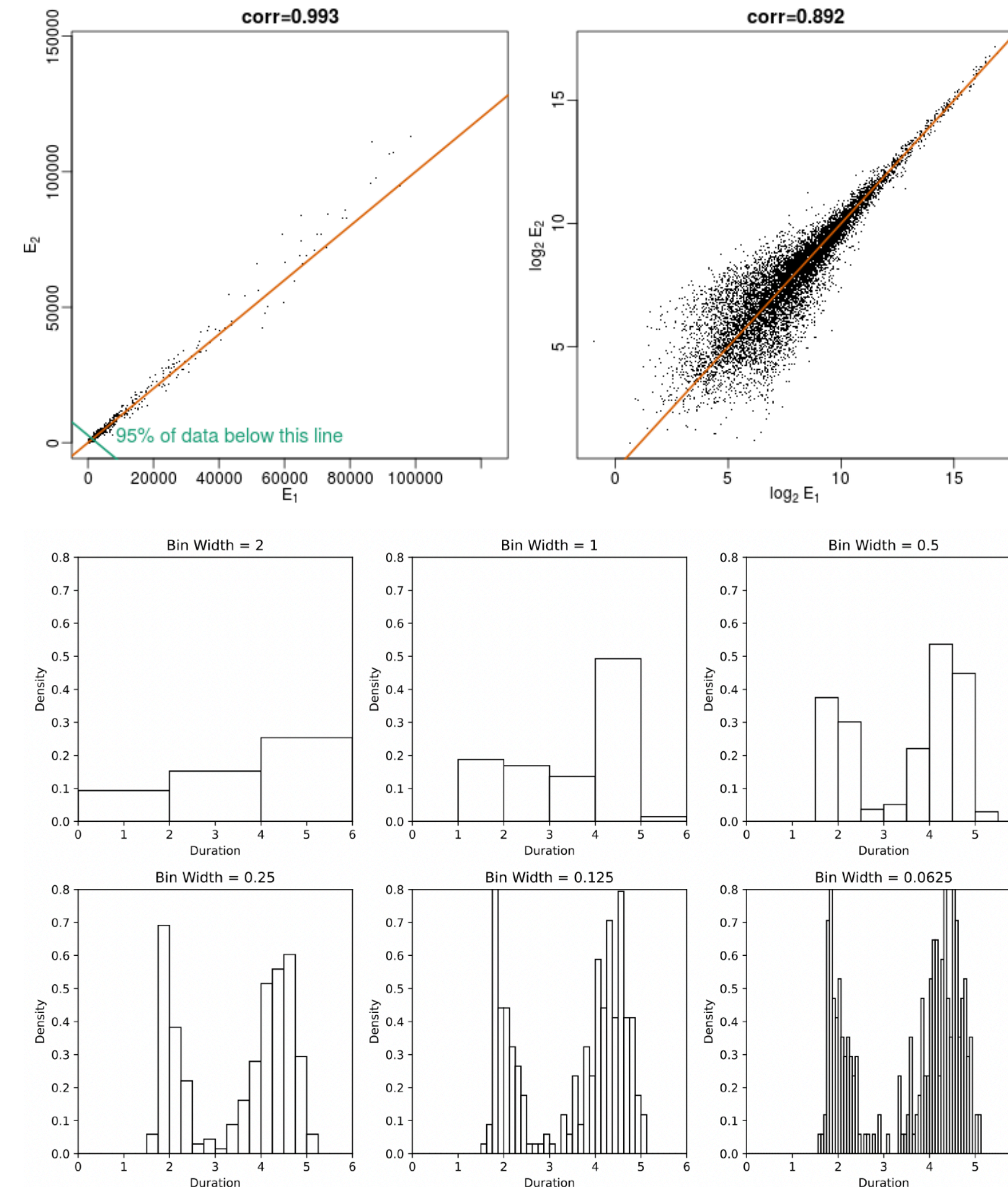


Plotting choices are scientific choices

Think about:

- axis limits
- linear vs log scales
- binning
- smoothing
- colour scales
- uncertainties
- comparison data
- selection effects / missing data

**A figure does not merely display the analysis.
It is part of the analysis.**



The 10-second rules I

Before using an important figure:

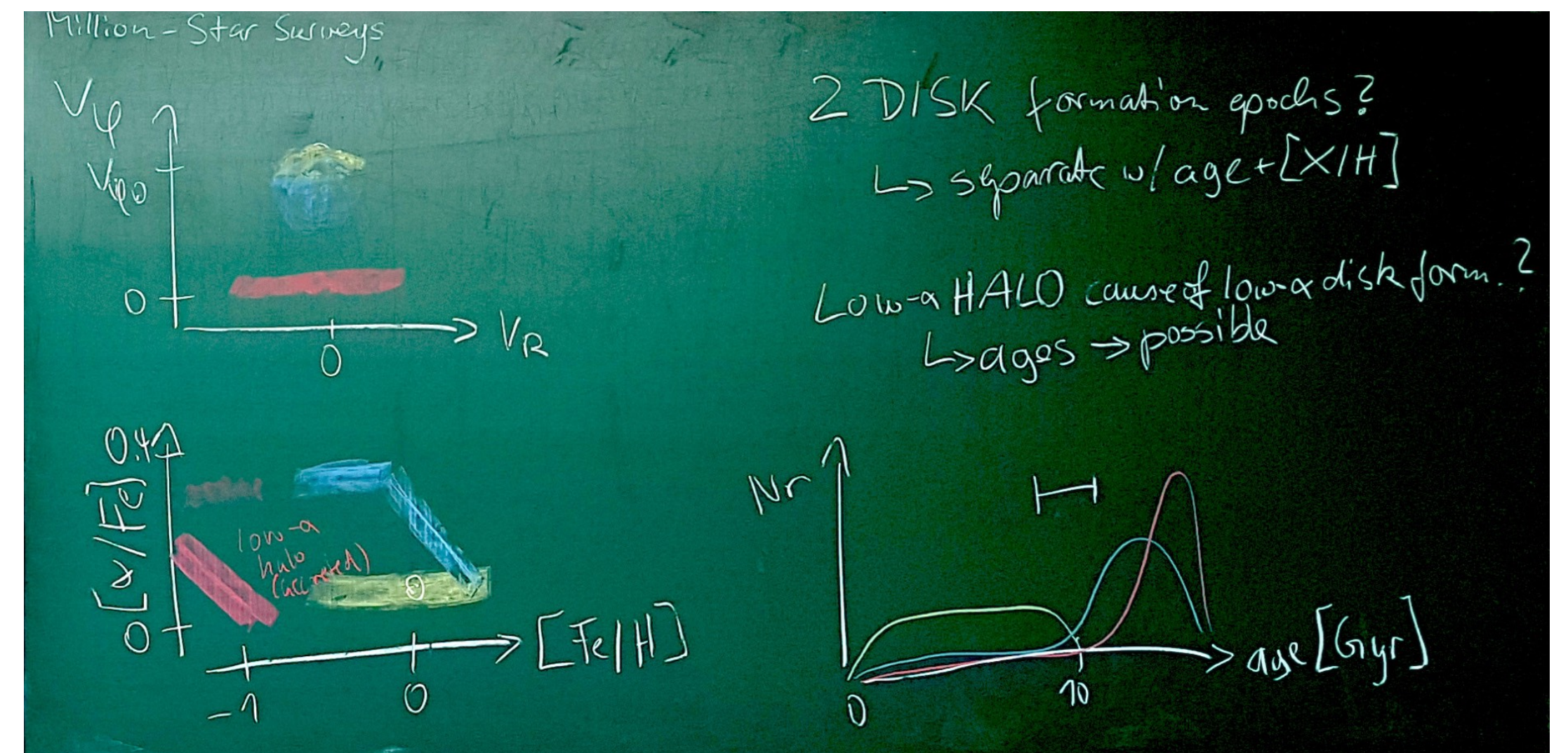
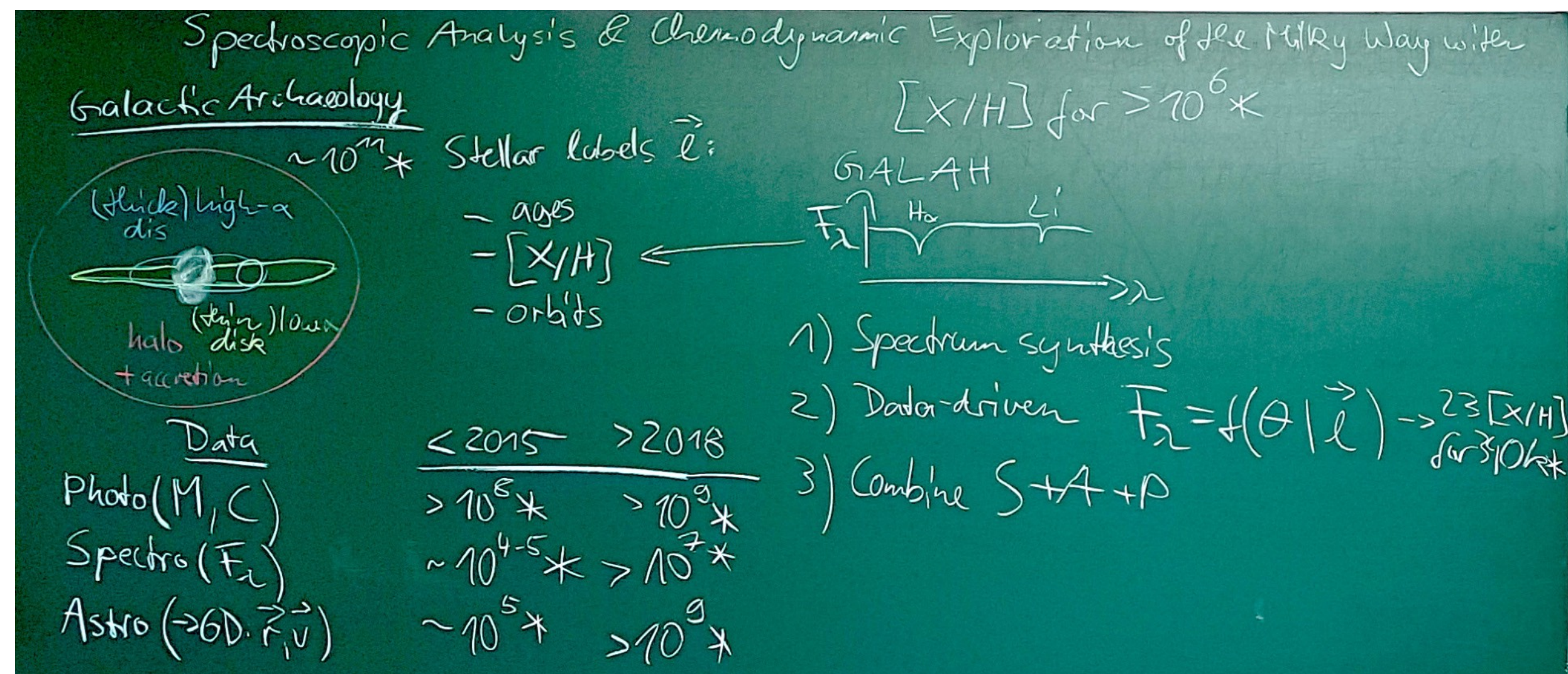
1. Show it to someone for **~10 seconds**
2. Hide it and ask:
 - > What was the figure about?
 - > What did you look at first?
 - > What do you remember?

If the answer is not what you intended...

...change the figure.

The 10-second rules II

Can you draw your figure in 10 seconds?



Examples of a PhD thesis defence of Heidelberg University:

20 minute blackboard talk with chalk (no erasing)

Before making your next plot...

Ask:

What should they SEE?

Where should the eye go first?

What should they UNDERSTAND?

What scientific point does the plot establish?

What should they REMEMBER?

What should still be in their head tomorrow?

Plotting all your data...

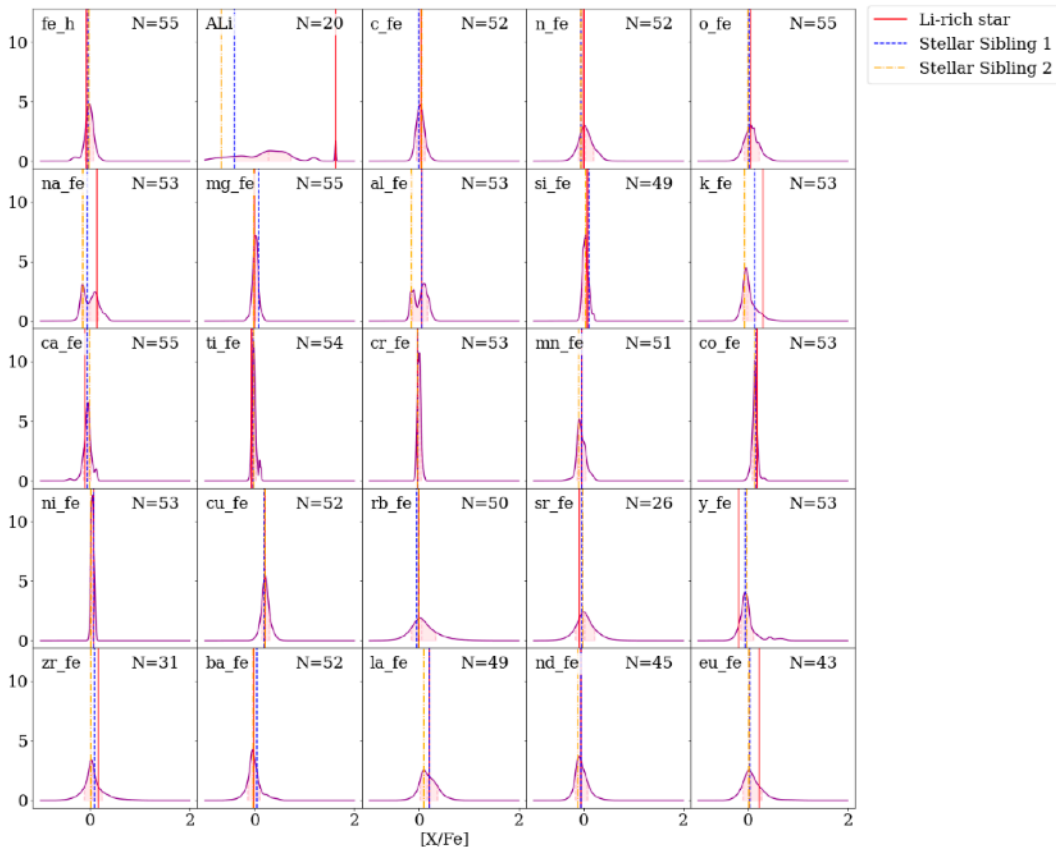
...is not the same as making a figure.

Make it as simple as the message allows.

Manage complexity.
(minimise tables/equations)

Design for your audience.

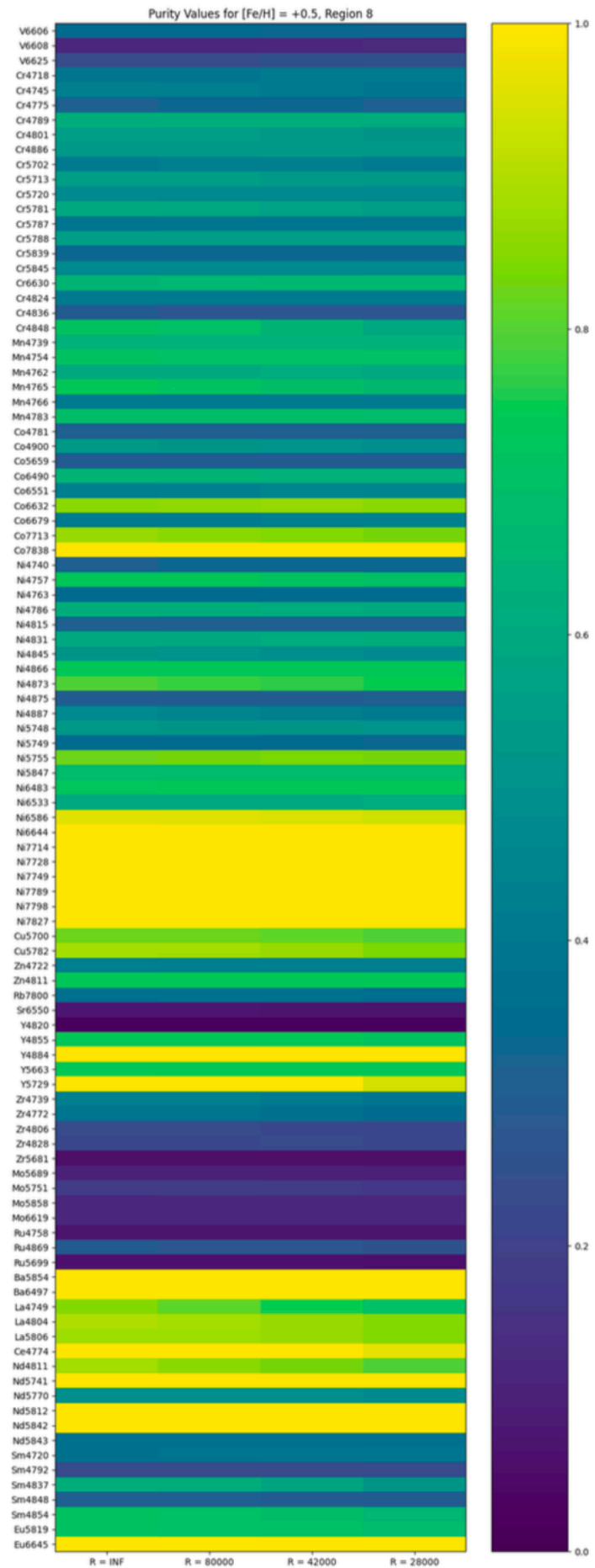
Make the science memorable.



	vmic	e_vmic	vsini	e_vsini	Type
0	1.695049	0.281026	4.033866	1.428236	Li-rich
1	1.499486	0.280871	3.999908	1.438448	Twin 1
2	1.654883	0.280399	1.396419	1.484546	Twin 2

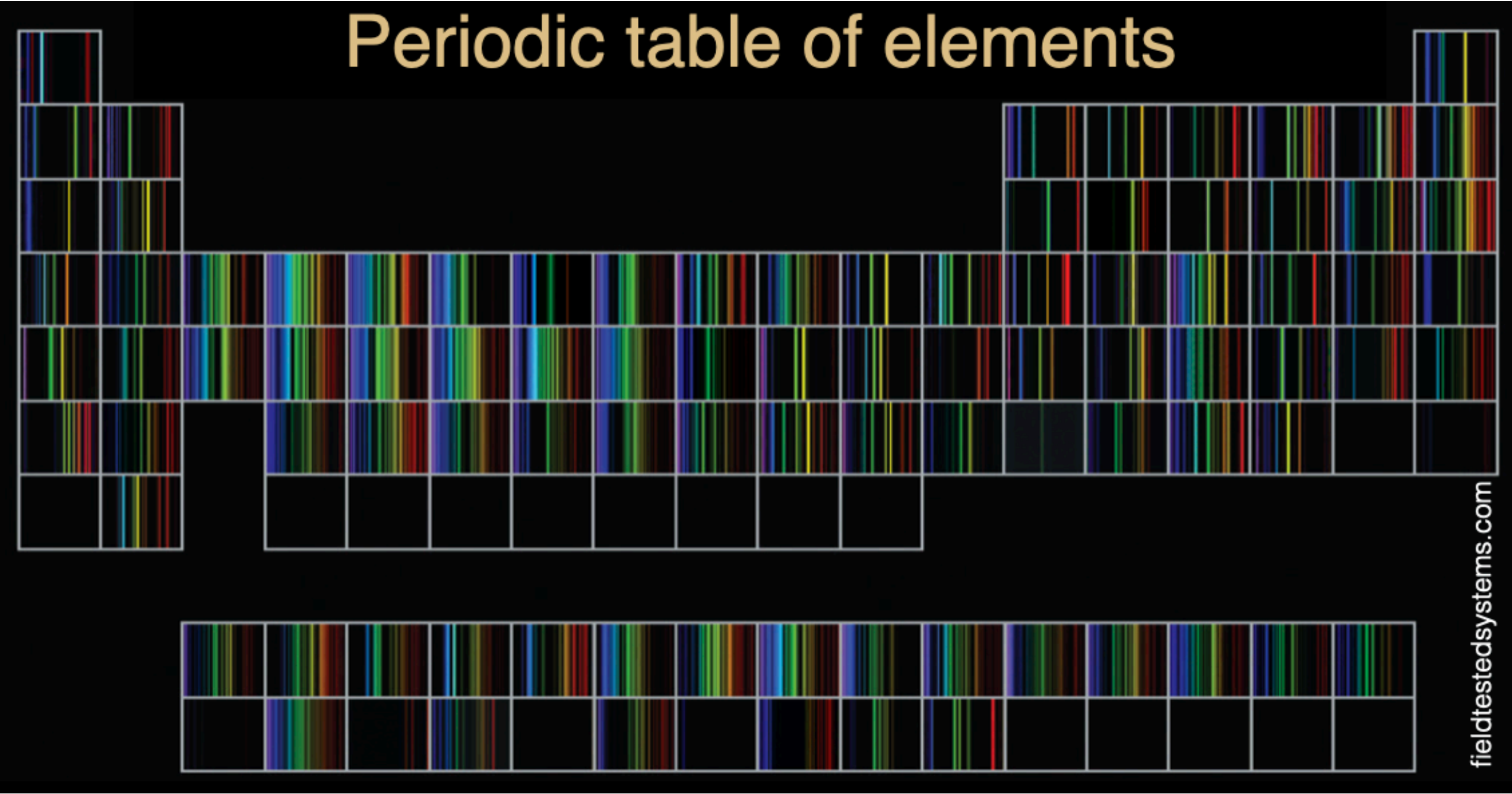
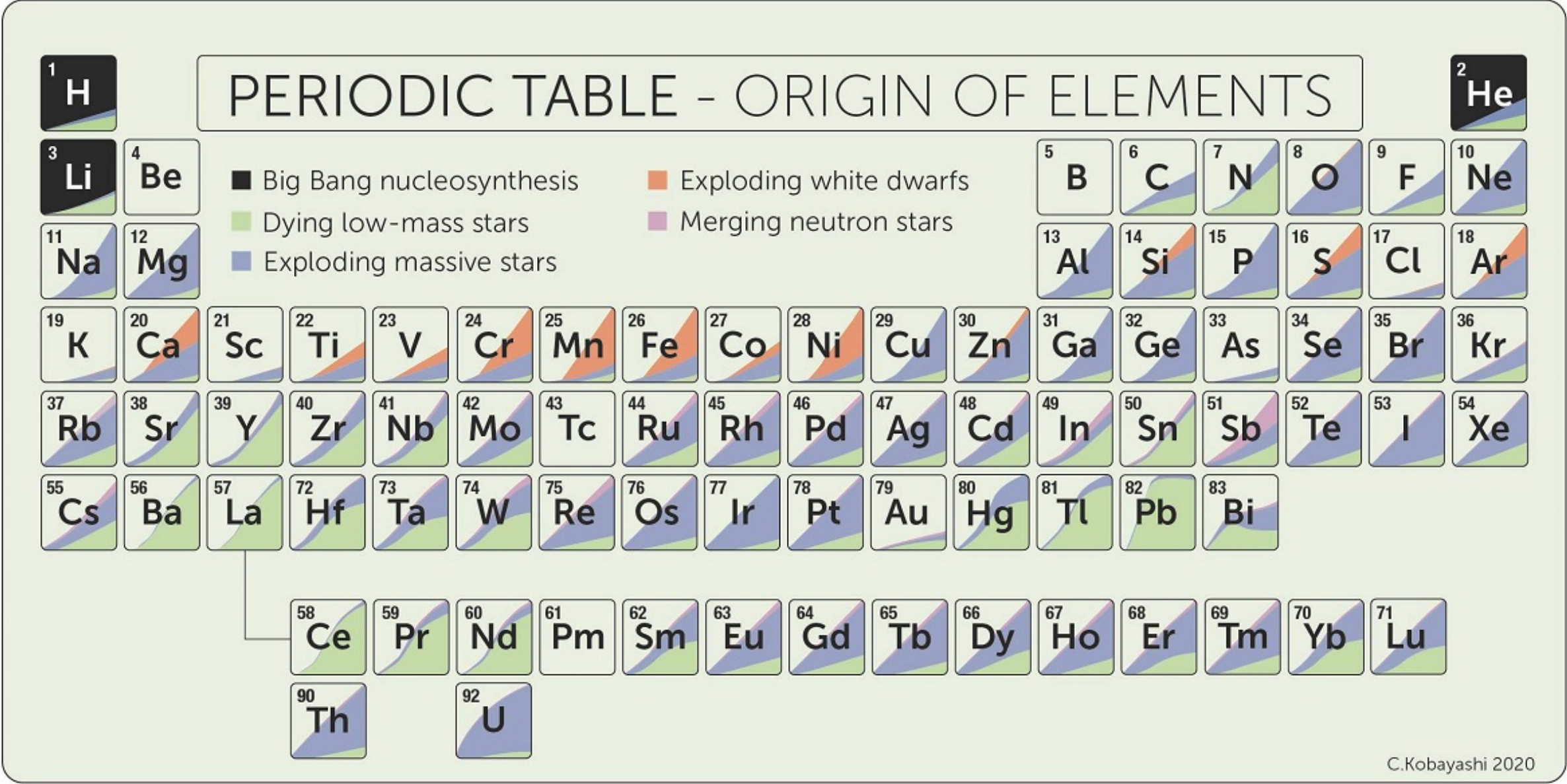
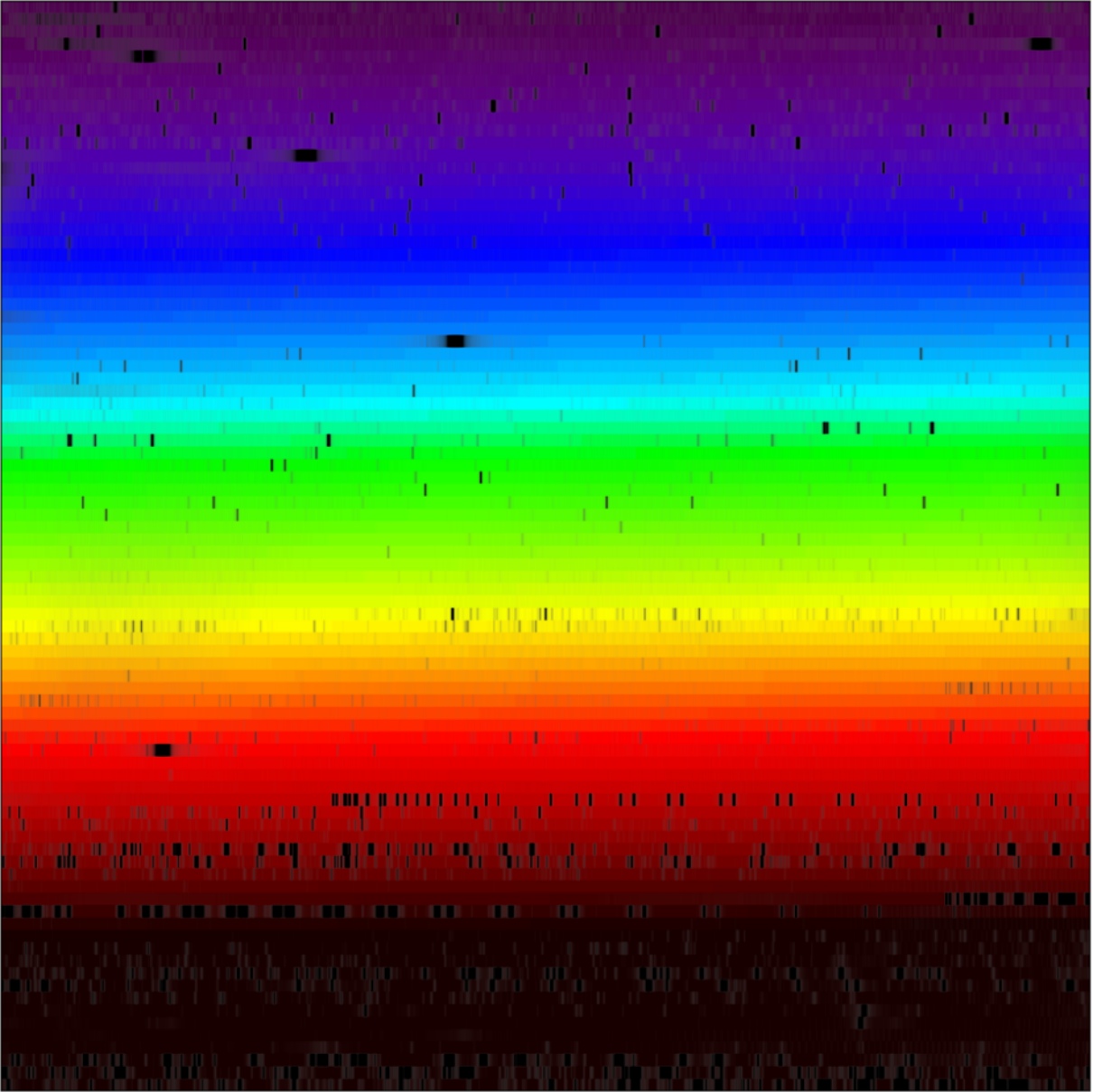
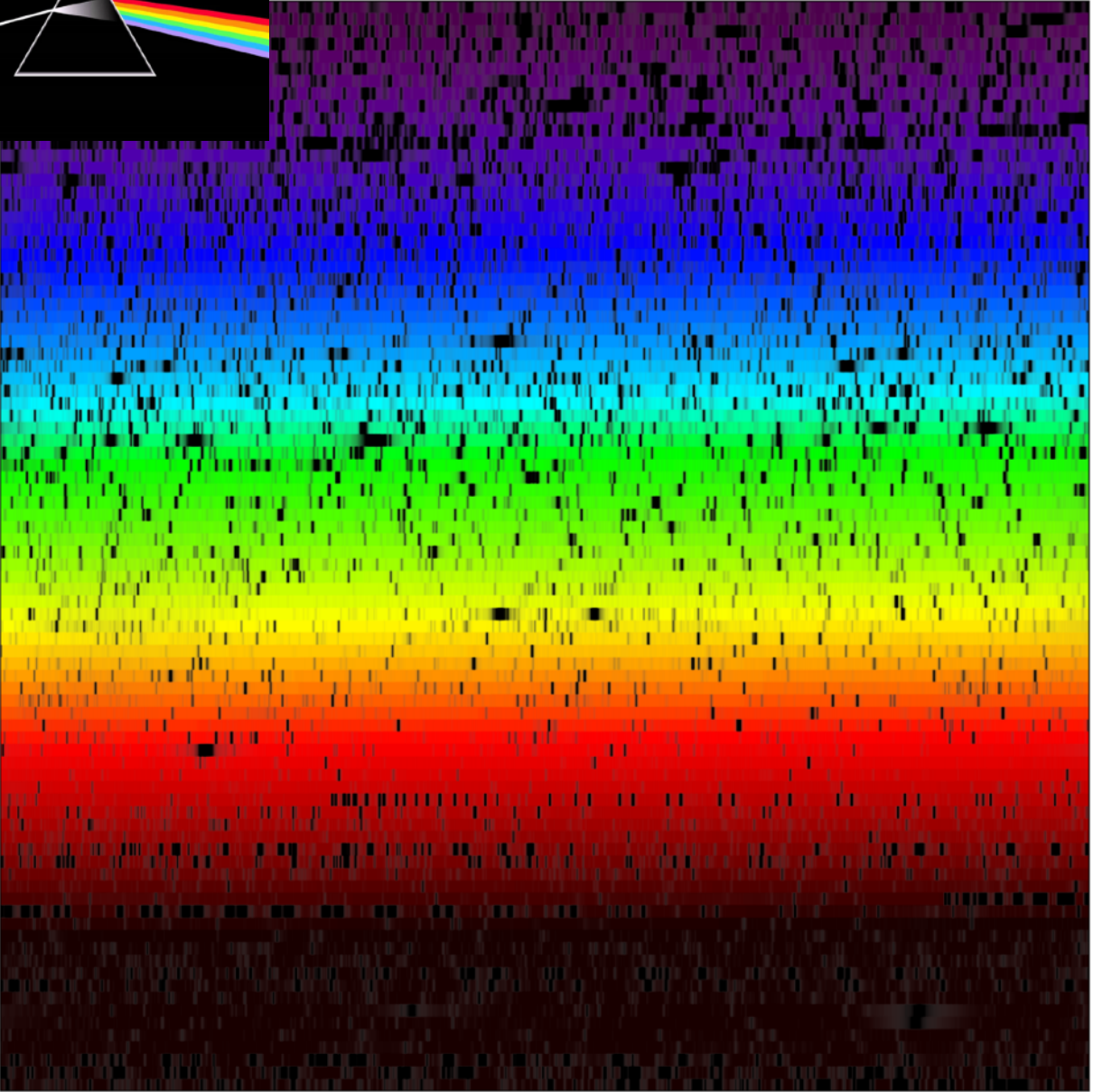
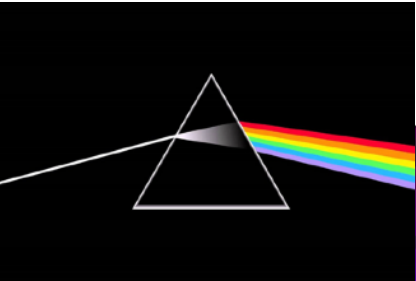
	ew_h_alpha	res_h_alpha	ew_h_beta	res_h_beta	Type
0	0.629786	1.167238	0.233924	0.848521	Li-rich
1	0.580490	1.100568	0.257781	0.853518	Twin 1
2	0.580168	1.144786	0.279170	0.908435	Twin 2

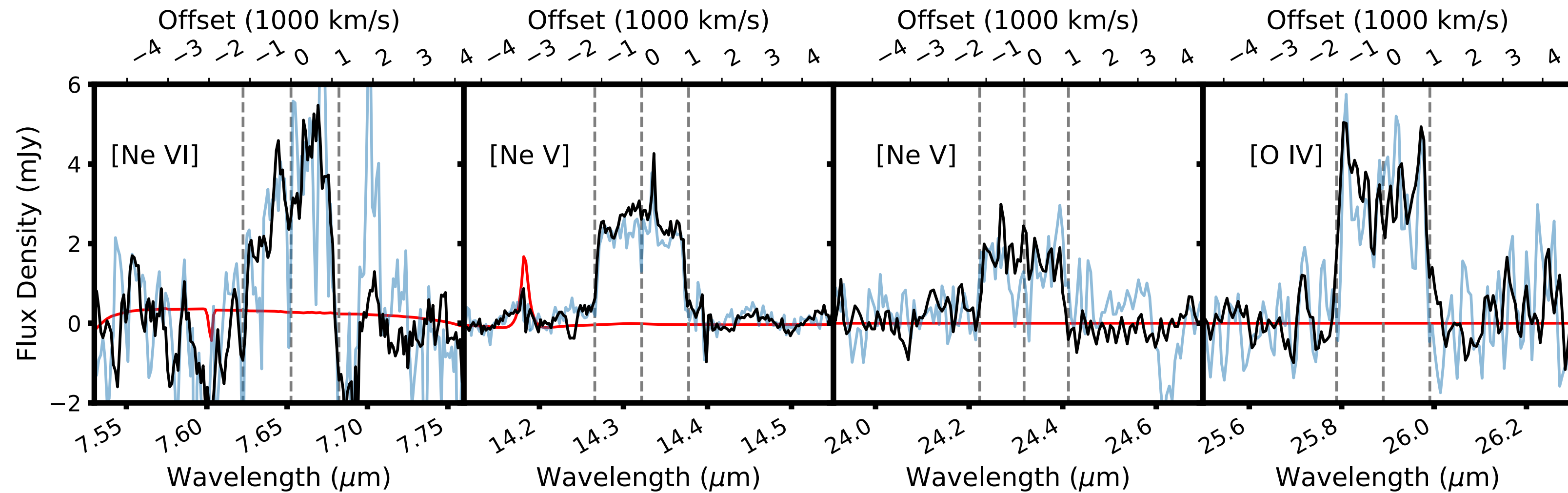
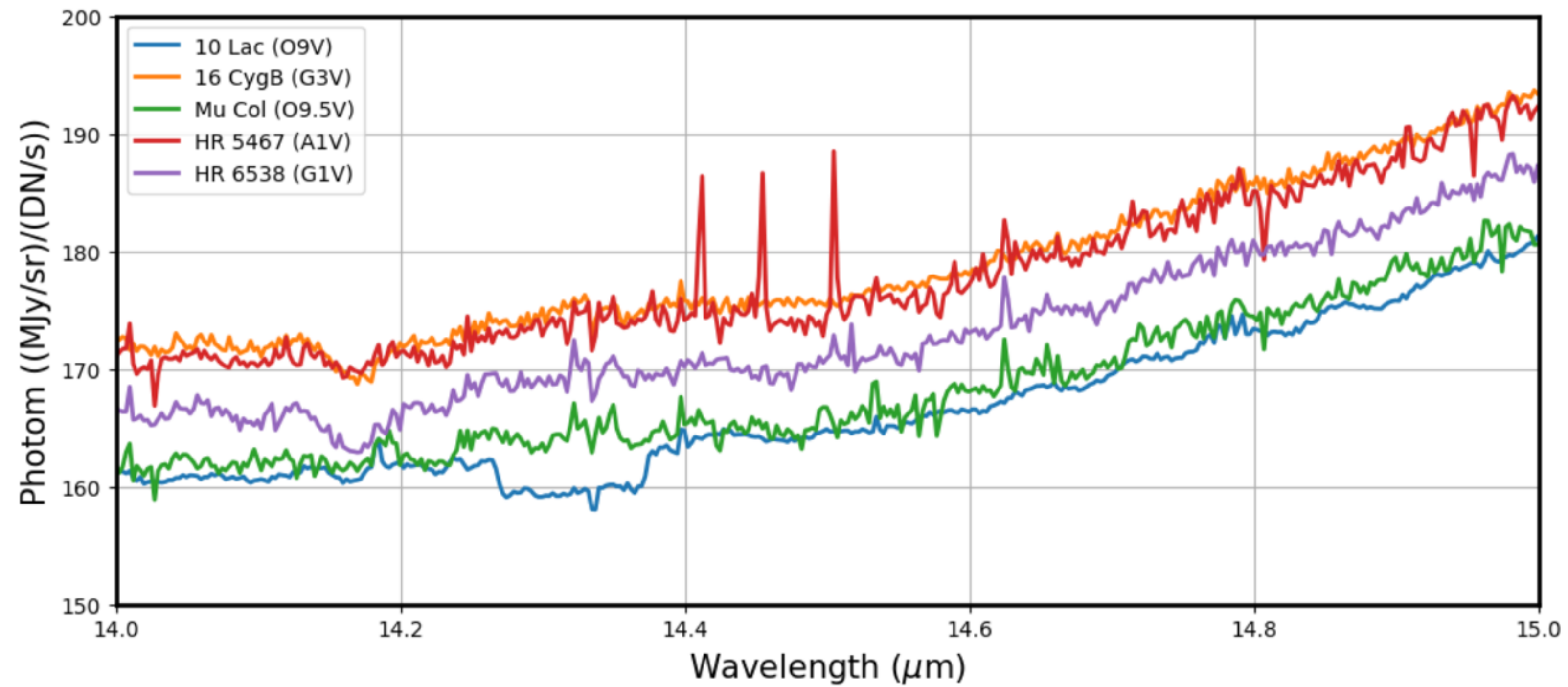
	k_fe	e_k_fe	ew_k_is	sigma_k_is	rv_k_is	Type
0	0.294419	0.048667	0.057230	0.130959	32.918465	Li-rich
1	0.135972	0.039312	0.051436	0.129009	34.493393	Twin 1
2	-0.068424	0.030965	0.017562	0.114227	32.623077	Twin 2



1A																		18 VIII A	
1 H Hydrogen 1.008																		2 He Helium 4.002602	
3 Li Lithium 6.94	4 Be Beryllium 9.0121831											5 B Boron 10.81	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998403163	10 Ne Neon 20.1797		
11 Na Sodium 22.98976928	12 Mg Magnesium 24.305											13 Al Aluminium 26.9815385	14 Si Silicon 28.085	15 P Phosphorus 30.973761998	16 S Sulfur 32.06	17 Cl Chlorine 35.45	18 Ar Argon 39.948		
19 K Potassium 39.0983	20 Ca Calcium 40.078	21 Sc Scandium 44.955908	22 Ti Titanium 47.867	23 V Vanadium 50.9415	24 Cr Chromium 51.9961	25 Mn Manganese 54.938044	26 Fe Iron 55.845	27 Co Cobalt 58.933194	28 Ni Nickel 58.6934	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.630	33 As Arsenic 74.921595	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798		
37 Rb Rubidium 85.4678	38 Sr Strontium 87.62	39 Y Yttrium 88.90584	40 Zr Zirconium 91.224	41 Nb Niobium 92.90637	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.90550	46 Pd Palladium 106.42	47 Ag Silver 107.8682	48 Cd Cadmium 112.414	49 In Indium 114.818	50 Sn Tin 118.710	51 Sb Antimony 121.760	52 Te Tellurium 127.60	53 I Iodine 126.90447	54 Xe Xenon 131.293		
55 Cs Caesium 132.90545196	56 Ba Barium 137.327	57 - 71 Lanthanoids		72 Hf Hafnium 178.49	73 Ta Tantalum 180.94788	74 W Tungsten 183.84	75 Re Rhenium 186.207	76 Os Osmium 190.23	77 Ir Iridium 192.217	78 Pt Platinum 195.084	79 Au Gold 196.966569	80 Hg Mercury 200.592	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98040	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)	
87 Fr Francium (223)	88 Ra Radium (226)	89 - 103 Actinoids		104 Rf Rutherfordium (267)	105 Db Dubnium (268)	106 Sg Seaborgium (269)	107 Bh Bohrium (270)	108 Hs Hassium (269)	109 Mt Meitnerium (278)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (289)	116 Lv Livermorium (293)	117 Ts Tennessee (294)	118 Og Oganesson (294)	

57 La Lanthanum 138.90547	58 Ce Cerium 140.116	59 Pr Praseodymium 140.90766	60 Nd Neodymium 144.242	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.964	64 Gd Gadolinium 157.25	65 Tb Terbium 158.92535	66 Dy Dysprosium 162.500	67 Ho Holmium 164.93033	68 Er Erbium 167.259	69 Tm Thulium 168.93422	70 Yb Ytterbium 173.045	71 Lu Lutetium 174.9668
89 Ac Actinium (227)	90 Th Thorium 232.0377	91 Pa Protactinium 231.03588	92 U Uranium 238.02891	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (260)





Law et al. (2024): JWST MIRI MRS photometric calibration vectors from various stars.